Yale Invitational - Repaired practice edition - Answer key

Division C | Forensics | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

I-4. (2 points) Authored replacement for source item Question I-4: Burnt-hair odor plus crushable black bead and scales suggests:

Unit: Fibers: Classification and Microscopy.
Worked-case data: The questioned fiber is a flattened twisted ribbon with a cellulose-associated reference response. Reference A is cotton, B is wool, and C is a uniform synthetic filament.
Separate investigation observations:
Longitudinal view: A flattened ribbon with repeated twists is visible.
Composition record: The supplied reference assay supports a cellulose-based material.
Comparison set: Wool shows scales; the synthetic reference is a smooth uniform filament; cotton shows convolutions.

A. Polyester

B. Wool

C. Cotton

D. Acrylic

Correct: B